

OWASP AppSecEU 2018 – Attacking "Modern" Web Technologies

OWASP AppSecEU 2018 – Attacking "Modern" Web Technologies - Frans Rosén, Slideshare.

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[image: Attacking Modern Web Technologies Frans Rosén @fransrosen]

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[image: Modern = stuff people use Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt.]

[image: Author name her Frans Rosén Attacking Modern Web Technologies Frans Rosén @fransrosen • "The Swedish Ninja" • Security Advisor @detectify (twitter: @fransrosen) • HackerOne #7 @ /leaderboard/all-time • Blog at labs.detectify.com]

[image: Author name her Frans Rosén Attacking Modern Web Technologies Frans Rosén @fransrosen • Winner of MVH at H1-702 Live Hacking in Vegas! • Winner Team Sweden in San Francisco (Oath) • Best bug at H1-202 in Washington (Mapbox) • Best bug at H1-3120 in Amsterdam (Dropbox)]

![Attacking Modern Web Technologies Rundown AppCache • Bug in all browsers

Upload Policies • Weak Implementations • Bypassing business logic

Deep dive in postMessage implementations • The postMessage-tracker extension • Abusing sandboxed domains • Leaks, extraction, client-side race conditions Frans Rosén @fransrosen]

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![Attacking Modern Web Technologies Rundown Frans Rosén @fransrosen Tool share! AppCache • Bug in all browsers

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[image: Author name her Disclaimer Attacking Modern Web Technologies Frans Rosén @fransrosen <https://speakerdeck.com/filedescriptor/exploiting-the-unexploitable-with-lesser-known-browser-tricks?slide=22> Found independently by @filedescriptor Announced last AppSecEU]

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[image: Author name her Cookie Stuffing/Bombing Attacking Modern Web Technologies Frans Rosén @fransrosen Will make EVERY page return 500 Error = Manifest FALLBACK will be used]

[image: Author name her Bug in every browser Attacking Modern Web Technologies Frans Rosén @fransrosen Manifest placed in /u/2241902/manifest.txt Would use the FALLBACK for EVERYTHING, even outside the dir]

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[image: image]

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[image: Author name her AppCache vulns still possible Attacking Modern Web Technologies Frans Rosén @fransrosen Requirements: • HTTPS only (was changed recently) • Files uploaded can run HTML • Files could be on a isolated sandboxed domain • Files are uploaded to the same directory for all users]

[image: Author name her ServiceWorkers, big brother of AppCache Attacking Modern Web Technologies Frans Rosén @fransrosen Requirements: • HTTPS only • Files uploaded can run HTML • Files could be on a isolated sandboxed domain • Files are uploaded to the root path For example: bucket123.s3.amazonaws.com/test.html]

[image: Upload Policies AWS and Google Cloud Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit

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We can place stuff in root, remember ServiceWorkers/AppCache?]

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We can still upload HTML: Content-type: image/jpeg;text/html]

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[image: Author name her Vulnerabilities Attacking Modern Web Technologies Frans Rosén @fransrosen " We can select what file to override]

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Just fetch the URL and we have the invoice POST /user_uploads/signed_url/ HTTP/1.1 Host:

example.com Content-Type: application/json;charset=UTF-8

```
{"file_name": "documents/invoice1.pdf", "content_type": "application/pdf"}
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```
{"signed_url": "https://storage.googleapis.com/uploads/documents/invoice1.pdf?"
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```
Expires=1515198382&GoogleAccessId=example%40example.iam.gserviceaccount.com&
```

```
Signature=dIMAFC2Gs22eP%2ByoAhwGqo0A0ijySYtRdkaIHVUr%2FvwKfNSKkKwTTPbpyOF..."}
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[image: Rolling your own policy logic sucks Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt.]

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Trigger: { "params": { "testRules": { "rules": [{ "name": "xxx", "triggers": { "type": "Delay", "delay": 5000 } ...
}] } } }](https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-72-320.jpg)

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"type": "JSVariableExists", "name": "ClickTaleCookieDomain", "value": "example.com" }, ...

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Action: ... "action": { "actualType": "CTEventAction", "type": "TestRuleEvent", "dynamicEventName": {
"parts": [{ "type": "ElementValue", "ctSelector": { "querySelector": ".content-wrapper script" } }, { "type":
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(https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-98-320.jpg)

![setInterval(function() { if(b) b.postMessage({"sitelist":"www.example.com/global","siteurl":"www.example.com/uk","curr":"curr=&osl='- (function() {document.body.appendChild(iframe=document.createElement('iframe'));window.alert=iframe.contentWindow['alert'];document.body.removeChild(iframe);window.alert(document.domain)})()-''','*') }, 10); Author name her Loaded JS Attacking Modern Web Technologies Frans Rosén @fransrosen link.com.example.com Loads mpel.js... Loading...]
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